

Network Security & Scanning

Objective

The objective of this task was to gain practical experience in reconnaissance, network scanning, vulnerability assessment, packet analysis, and firewall configuration. The task involved identifying services running on a target machine, discovering vulnerabilities, analyzing network traffic, and implementing basic security controls.

1. Reconnaissance

Reconnaissance is the process of gathering information about a target before performing security assessments. It helps identify potential attack surfaces and provides valuable information about systems and services.

WHOIS Lookup

WHOIS was used to obtain domain registration information such as ownership details, registrar information, and registration dates.

Purpose:

- Gather publicly available domain information.
- Understand domain ownership and infrastructure details.

NSLOOKUP

NSLOOKUP was used to query DNS records and obtain information about domain name resolution.

Purpose:

- Identify IP addresses associated with a domain.
- Examine DNS records and configurations.

Google Dorking

Google Dorking uses advanced search operators to discover publicly accessible information that may not be easily found through normal searches.

Example:

site:example.com filetype:pdf

Purpose:

- Identify publicly exposed documents.

- Discover sensitive information indexed by search engines.

Shodan

Shodan is a search engine that indexes internet-connected devices and services.

Purpose:

- Discover exposed servers and services.
 - Understand how publicly accessible systems can be identified.
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2. Port and Service Scanning

Port scanning helps identify active services running on a target system. It is one of the most important steps in network enumeration.

TCP SYN Scan

Command:

```
nmap -sS 192.168.56.102
```

Purpose:

- Detect open TCP ports.
- Perform a stealth scan without completing the full TCP handshake.

Observation:

Several services running on the target machine were identified through open ports.

UDP Scan

Command:

```
nmap -sU 192.168.56.102
```

Purpose:

- Identify active UDP services.
 - Detect services that do not use TCP communication.
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Service Version Detection

Command:

```
nmap -sV 192.168.56.102
```

Purpose:

- Identify software versions running on detected services.
- Assist in vulnerability identification.

Observation:

Service versions can reveal outdated or vulnerable software.

Operating System Detection

Command:

```
nmap -O 192.168.56.102
```

Purpose:

- Estimate the operating system running on the target machine.
 - Support further vulnerability assessment.
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3. Vulnerability Assessment**Nessus Essentials**

Nessus Essentials was used to perform vulnerability scanning on the Metasploitable2 virtual machine.

Vulnerability Severity Levels

- Critical
- High
- Medium
- Low
- Informational

Purpose

- Identify security weaknesses.
- Evaluate the risk level of discovered vulnerabilities.
- Generate professional vulnerability reports.

Observation

The scan revealed multiple intentionally vulnerable services on Metasploitable2, making it suitable for cybersecurity training and testing.

4. Packet Analysis Using Wireshark

Wireshark is a network protocol analyzer used to inspect and monitor network traffic in real time.

HTTP Traffic Analysis

Captured HTTP packets and observed request-response communication.

Purpose:

- Understand web communication.
 - Examine unencrypted network traffic.
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DNS Traffic Analysis

Captured DNS packets and analyzed domain name resolution.

Purpose:

- Understand how hostnames are translated into IP addresses.
 - Monitor DNS queries and responses.
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FTP Credential Analysis

Captured FTP traffic and observed credential transmission.

Observation:

FTP sends credentials in plaintext, making it insecure for sensitive communication.

Purpose:

- Demonstrate the importance of secure protocols.
 - Understand risks associated with unencrypted traffic.
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5. SYN Flood Simulation

hping3

Command:

```
sudo hping3 -S 192.168.56.102 -p 80 -c 50
```

Purpose

- Simulate a SYN flood attack.
- Observe abnormal TCP connection requests.
- Understand denial-of-service attack behavior.

Observation

Multiple SYN packets were generated and observed using Wireshark. This demonstrated how attackers can attempt to overwhelm a target system by creating numerous incomplete TCP connections.

6. Firewall Configuration

iptables Rule Creation

Command:

```
sudo iptables -A INPUT -p tcp --dport 80 -j DROP
```

Purpose

- Block incoming traffic on port 80.
- Demonstrate basic firewall filtering.

Observation

After applying the firewall rule, network scans showed that access to the protected port was restricted.

Port Scan Verification

Command:

```
nmap -p 80 192.168.56.102
```

Observation

The scan indicated that the port was filtered or inaccessible, confirming that the firewall rule was functioning correctly.

Key Learning Outcomes

- Learned reconnaissance techniques used during information gathering.
 - Performed TCP and UDP scanning using Nmap.
 - Identified services and operating systems running on a target machine.
 - Conducted vulnerability assessment using Nessus Essentials.
 - Analyzed HTTP, DNS, and FTP traffic using Wireshark.
 - Simulated a SYN flood attack using hping3.
 - Configured firewall rules using iptables.
 - Improved understanding of network security monitoring and defensive techniques.
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Conclusion

This task provided practical exposure to reconnaissance, scanning, vulnerability assessment, packet analysis, attack simulation, and firewall configuration. The exercises helped build a deeper understanding of network security concepts and the tools commonly used by cybersecurity professionals for both offensive and defensive security operations.